05c — CDN (Content Delivery Network)

Key Concepts

- **CDN** — a network of geographically distributed servers that cache content close to users
- **Edge node / Edge server** — a CDN server in a specific geographic location
- **Origin server** — your actual backend server where content originates
- **Cache purging** — forcefully deleting cached content from CDN before TTL expires
- **Cache versioning** — changing the URL when content changes so CDN treats it as a new file

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How It Works

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Without CDN:  Mumbai User → Virginia Server (~13,000 km, ~200ms latency)

With CDN:     Mumbai User → CDN Edge Node in Mumbai (~5ms latency)
                                ↑
                         Cached origin content
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1. User requests content
2. Request hits nearest CDN edge node
3. **Cache hit** → served immediately from edge
4. **Cache miss** → edge fetches from origin, caches it, serves it
5. Next user in same region → cache hit

What CDNs Cache

- Static assets: images, CSS, JS, fonts, videos
- API responses (if `Cache-Control: public`)
- Entire HTML pages (static sites)
- **Not cached:** dynamic, personalized content (user feeds, dashboards)

CDN Cache Invalidation Strategies

Strategy	How	When to Use
TTL expiry	Cache auto-expires after set time	Routine updates
Cache purge	API call to delete specific cached files	Urgent updates
Cache versioning	Change URL when content changes (hash in filename)	Best practice for assets
Surrogate keys / Cache tags	Tag objects, purge by tag	One change affects many URLs

The rule: TTL handles routine expiry. Purging handles urgent updates. Versioning avoids the problem entirely.

Popular CDNs

CDN	Used By
Cloudflare	Millions of websites
AWS CloudFront	Amazon ecosystem
Akamai	Enterprise, streaming
Fastly	GitHub, Stripe, Reddit

Industry Examples

- **Netflix** — video files cached at edge nodes in 190+ countries; stream comes from ~50km away
- **Hotstar** — 25M concurrent IPL viewers served via CDN; origin server would have collapsed
- **GitHub** — JS/CSS on Fastly; fast globally despite single origin

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Interview Questions

Q: What is a CDN and why use one?

> A network of geographically distributed servers that cache content near users. Reduces latency and offloads traffic from the origin server.

Q: What happens when origin content changes?

> Cached copies remain until TTL expires. For urgent updates, use cache purging via CDN API. Best practice is cache versioning — change the URL when content changes.

Q: What does a CDN cache?

> Mainly static assets (images, JS, CSS, video). Dynamic or personalized content typically bypasses the CDN.

Q: What is cache versioning?

> Embedding a hash or version number in the asset URL. When the file changes, the URL changes — CDN treats it as a brand new file and fetches fresh from origin.

Q: What are surrogate keys?

> Cache tags that group related cached objects. Purging a tag clears all objects associated with it — useful when one DB change affects many cached URLs.